VINAY RAJU SANNIBOINA

Graduate Electrical Engineer — EV Powertrain, Power Electronics & Battery Systems

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PROFESSIONAL PROFILE

Electrical & Electronics Engineering graduate (BEng, First Class, Electric Vehicles specialisation) with an MSc in Automotive Engineering for Electric Vehicles. Grounded in electric powertrain fundamentals — electrical machines, power electronics, motor control and battery systems — and in simulation and analysis using MATLAB/Simulink and ANSYS. Hands-on site experience supporting the **installation, testing and commissioning of power transformers**, and CAE research into **battery thermal management for EV battery packs**. Seeking a UK graduate or junior engineering role in EV powertrain, power electronics, battery management systems or electrical testing and commissioning.

KEY SKILLS

Power Electronics & Machines: Power electronic converters and inverters, electrical machines, electric motor drives and motor control, electrical circuits and power systems

Battery Systems & BMS: EV battery systems, battery management system (BMS) principles, battery thermal management (BTMS), cell and pack thermal behaviour, coolant and heat-transfer analysis

Simulation & Modelling: MATLAB, Simulink, ANSYS Fluent, CAE, thermal and fluid analysis, numerical modelling, mesh independence studies, results validation

Testing & Commissioning: Power transformer installation support, testing and commissioning, transformer oil change and maintenance, site preparation, site health & safety and risk assessment

Control & Electronics: Control systems, embedded systems, automotive electronics

Tools & Design: SOLIDWORKS, technical drawings, Microsoft Excel, Word, PowerPoint

Professional: Cross-functional teamwork, technical report writing, problem solving, working to regulations, budgets and deadlines

EDUCATION

MSc Automotive Engineering for Electric Vehicles — 2:1 May 2025 – May 2026

University of Bedfordshire, Luton, United Kingdom

Key modules: Automotive Electronics & Control Systems; Engineering Analysis & Numerical Modelling; Thermodynamics, Heat Transfer & Fluid Dynamics; Vehicle Dynamics & Performance; Computational Fluid Dynamics.

BEng (Hons) Electrical & Electronics Engineering (Electric Vehicles) — First Class Jun 2020 – May 2024

Bharath Institute of Higher Education and Research, India

Key modules: Electrical Machines & Power Electronics; Electric Vehicle Technologies & Battery Systems; Electrical Circuits & Power Systems; Control Systems & Embedded Systems; Automotive Electronics.

ENGINEERING EXPERIENCE

CAE Engineer Intern (Research Internship) — Simulation Lab, Product Development Centre Nov 2024 – Jan 2025

Remote research internship — Battery Thermal Management Systems (BTMS) using nanofluids for EVs

- Investigated **battery thermal management systems using nanofluid coolants** for electric vehicle battery packs.
- Analysed coolant and heat-transfer behaviour in EV battery packs, applying thermal and fluid dynamics principles to keep cells within safe operating temperatures.
- Produced and presented research findings on nanofluid coolant performance, linking simulation-based analysis to practical EV battery thermal management design decisions.

Junior Site Supervisor (Internship) — Shri Abhirami Engineering Works Pvt Ltd, Chennai, India Jan 2024 – Jun 2024

Power transformer installation, testing and commissioning

- Prepared sites ahead of **power transformer** delivery, carrying out pre-arrival site set-up so installation could start on schedule.
- Supported the **installation, testing and commissioning** of power transformers alongside senior engineers.
- Carried out **transformer oil change and maintenance** work.
- Worked to site safety procedures and risk controls on industrial electrical sites.

Formula Student Team Member — Team Bedfordshire Racing, Luton, UK Sep 2025 – Jun 2026

Class II vehicle project — vehicle build, manufacture and quality assurance

- Worked within a **24-member multi-disciplinary team** across design, manufacturing and vehicle build, coordinating work to project milestones.

- Manufactured and assembled vehicle components (SOLIDWORKS CAD, cutting, welding, fitment) and checked parts against drawings and Formula Student technical regulations.
- Saved **over £500 against the build budget** through cost-conscious material and component decisions; worked to workshop safety procedures and risk assessments.

ENGINEERING PROJECTS

MSc Dissertation — CFD Simulation & Optimisation of a Multi-Element Front Wing Completed May 2026
University of Bedfordshire | ANSYS Fluent, SpaceClaim, SOLIDWORKS

- Achieved an approximate **45% reduction in drag** through iterative simulation-driven design optimisation.
- Ran mesh independence studies, selected turbulence models and validated results through pressure and velocity analysis; delivered a full technical report with design recommendations.

TRAINING & ADDITIONAL INFORMATION

Artificial Intelligence programme: IIT Kanpur (1Stop), Jun – Aug 2021 — two-month programme, certificate of participation.
Membership: Institution of Mechanical Engineers (IMechE)
Languages: English, Telugu